

Chapter 2, Problem 1PP

Problem

The essential component of a toaster is an electrical element (a resistor) that converts electrical energy to heat energy. How much current is drawn by a toaster with resistance $15\ \Omega$ at $110\ \text{V}$?

Step-by-step solution

Step 1 of 1

The essential component of a toaster is an electrical element (a resistor) that converts electrical energy to heat energy.

The value of the resistor, $R = 15\ \Omega$

The value of voltage, $v = 110\ \text{V}$

Write Ohm's law.

$$R = \frac{v}{i}$$

$$\begin{aligned} i &= \frac{v}{R} \\ &= \frac{110}{15} \\ &= 7.333\ \text{A} \end{aligned}$$

Therefore, the current drawn by the toaster is $7.333\ \text{A}$.